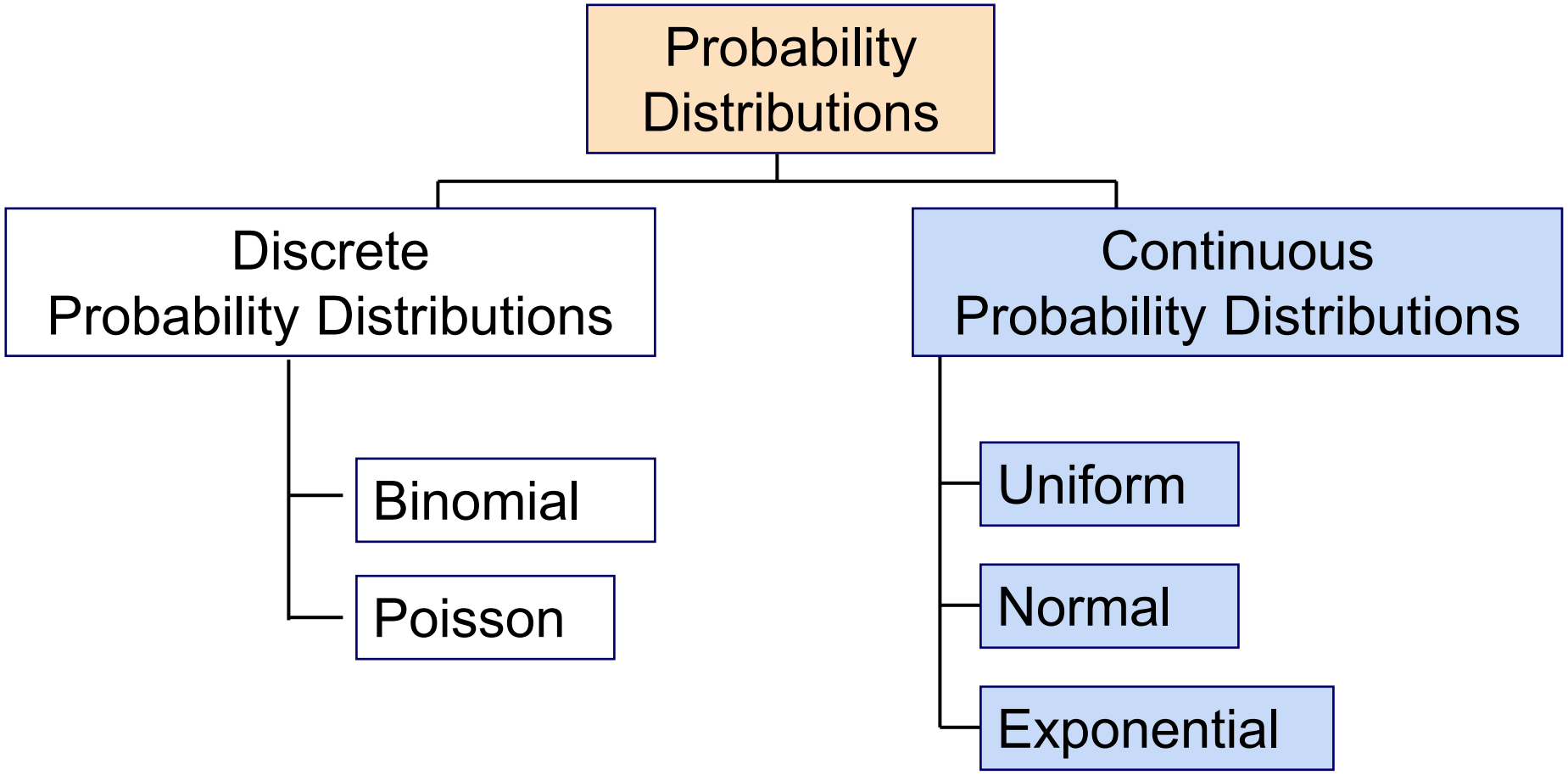
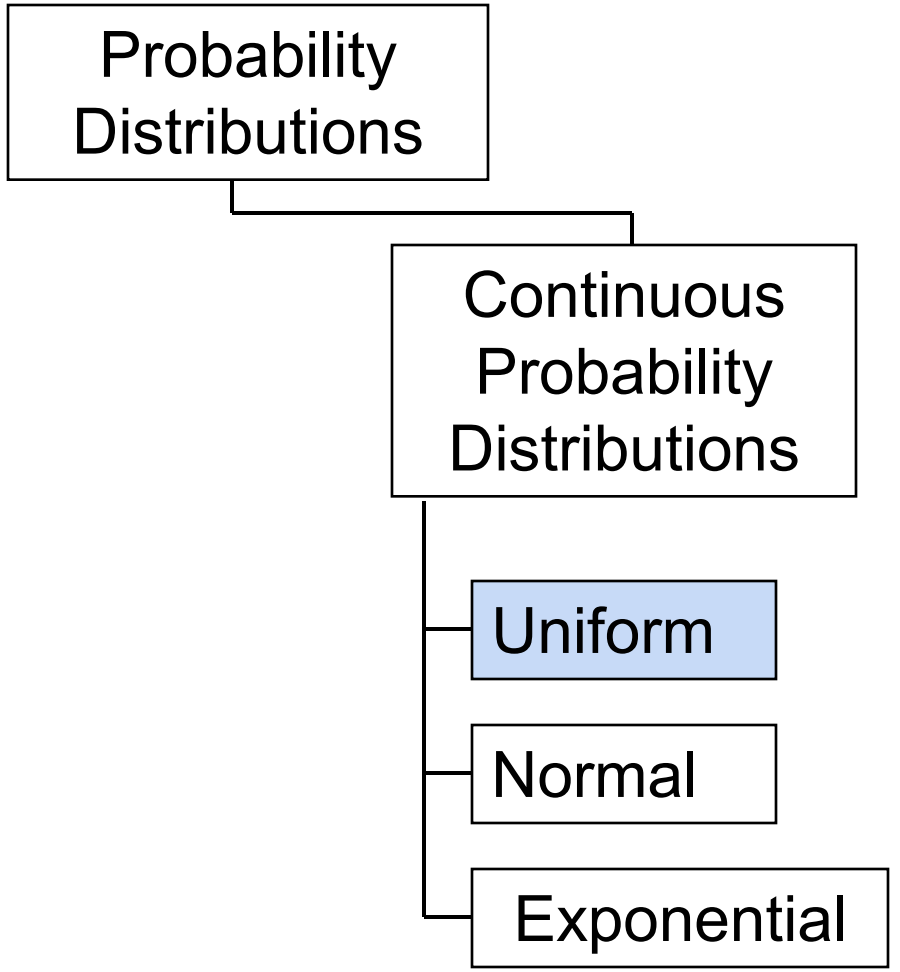
Continuous Probability Distribution

Statistics and Probability –MAS2001

Probability Distributions



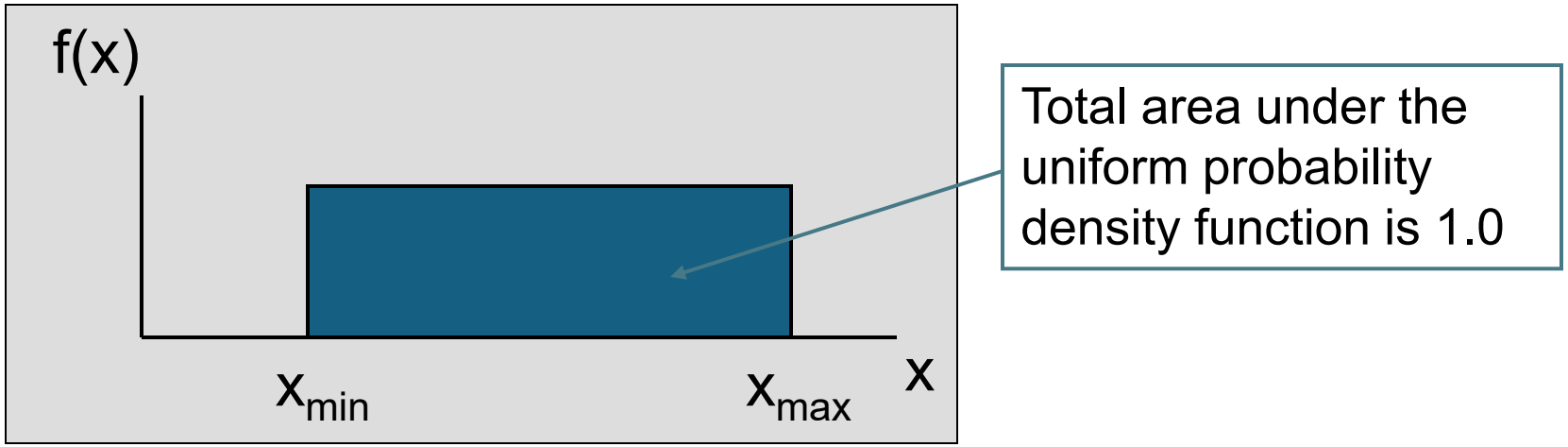
The Uniform Distribution



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The Uniform Distribution

- The **uniform distribution** is a probability distribution that has **equal probabilities** for all possible outcomes of the random variable



The Uniform Distribution

(continued)

The Continuous Uniform Distribution:

$$f(x) = \begin{cases} \dfrac{1}{b-a} & \text{if } a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

where

f(x) = value of the density function at any x value

a = minimum value of x

b = maximum value of x

Properties of the Uniform Distribution

- The mean of a uniform distribution is

$$\mu = \frac{a+b}{2}$$

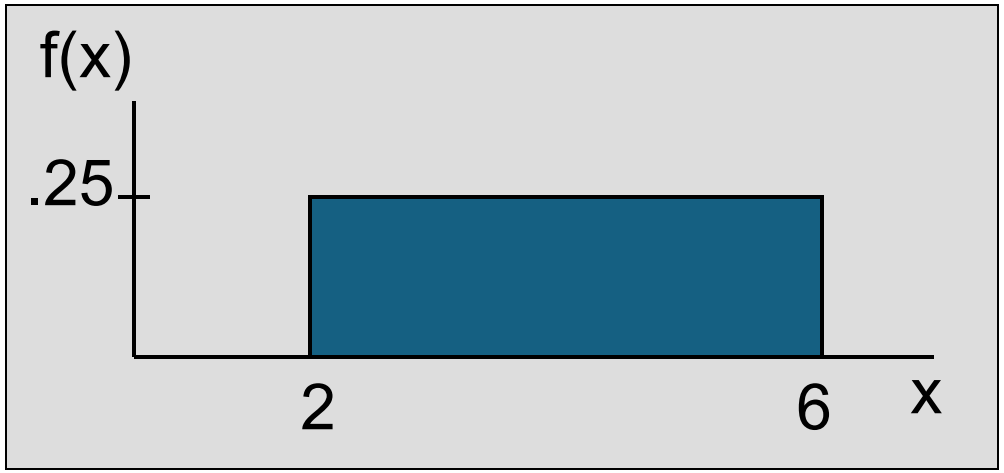
- The variance is

$$\sigma^2 = \frac{(b-a)^2}{12}$$

Uniform Distribution Example

Example: Uniform probability distribution over the range $2 \le x \le 6$:

$$f(x) = \frac{1}{6 - 2} = .25 \quad \text{for} \quad 2 \le x \le 6$$



$$\mu = \frac{a+b}{2} = \frac{2+6}{2} = 4$$

$$\sigma^2 = \frac{(b-a)^2}{12} = \frac{(6-2)^2}{12} = 1.333$$

Uniform distribution

The distribution function is given by:

$$F(x) = \begin{cases} 0 & x < a \\ \dfrac{x-a}{b-a} & a \leq x \leq b \\ 1 & x > b \end{cases}$$

A continuous random variable X is uniformly distributed on [0,1].

Find $P(0.2<X<0.8)$.

$P(0.2<X<0.8)$

$= F(0.8) - F(0.2)$

$= \frac{0.8-0}{1-0} - \frac{0.2-0}{1-0}$

$= 0.8 - 0.2 = 0.6$

The time (in minutes) taken by a student to complete a test is uniformly distributed between 30 and 90 minutes.
Find (i) the probability that the student completes the test in less than 45 minutes, (ii) the probability that the time lies between 50 and 70 minutes.

Let **X** be the continuous random variable representing the test completion time.

X follows a continuous uniform distribution: **X ~ U(a, b)**

Probability Density Function (PDF):

f(x) = 1 / (b - a) = 1 / 60 [for 30 ≤ x ≤ 90]

(i) Probability of completion in less than 45 minutes

P(X < 45) = (45 - a) / (b - a)

P(X < 45) = (45 - 30) / (90 - 30)

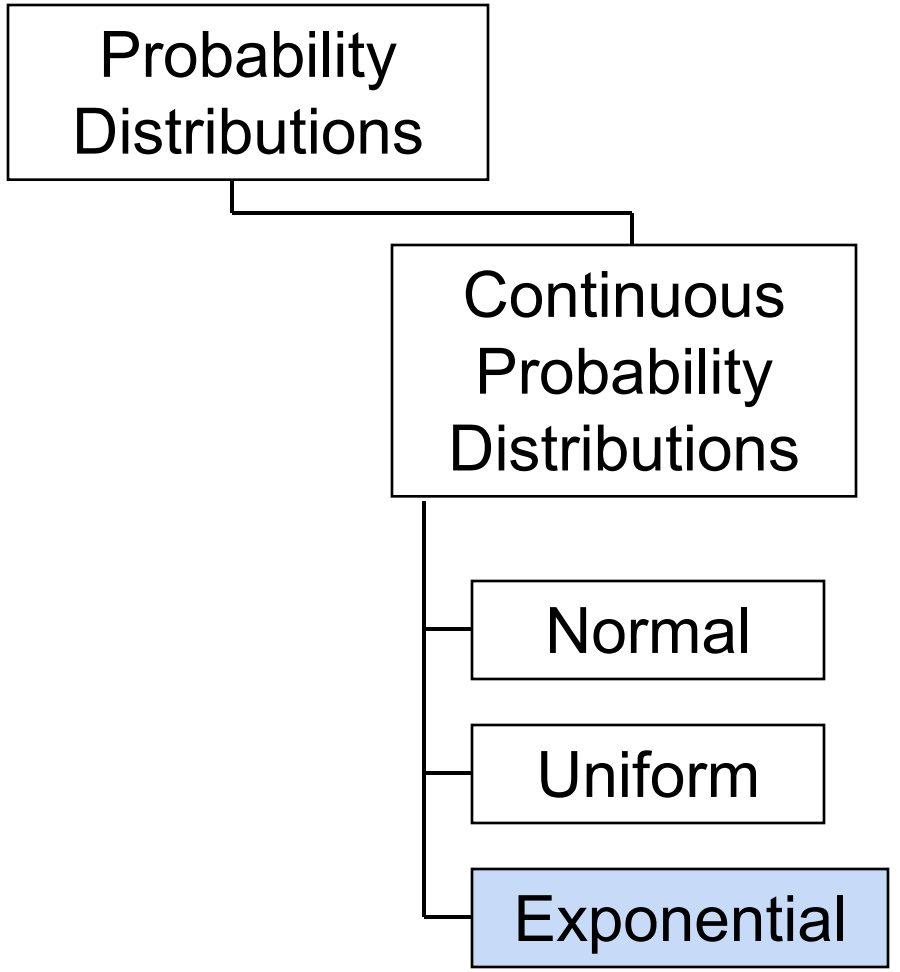
P(X < 45) = 15 / 60 = **0.25 (or 25%)**

(ii) Probability that the time lies between 50 and 70 minutes

P(50 < X < 70) = (70 - 50) / (90 - 30)

P(50 < X < 70) = 20 / 60 = 1 / 3 ≈ **0.3333 (or 33.33%)**

The Exponential Distribution



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The Exponential Distribution

- Used to model the length of time between two occurrences of an event (the time between arrivals)
  - Examples:
    - Time between trucks arriving at an unloading dock
    - Time between transactions at an ATM Machine
    - Time between phone calls to the main operator

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The Exponential Distribution *(continued)*

- The exponential random variable T (t>0) has a probability density function

$$f(t) = \lambda e^{-\lambda t} \quad \text{for } t > 0$$

- Where
  - λ is the mean number of occurrences per unit time
  - t is the number of time units until the next occurrence
  - e = 2.71828

The Exponential Distribution

- T is said to follow an exponential probability distribution

defined by a single parameter, its mean λ (lambda)

- The cumulative distribution function (the probability that an arrival time is less than some specified time t) is

$$F(t) = 1 - e^{-\lambda t}$$

where e = mathematical constant approximated by 2.71828

λ = the population mean number of arrivals per unit

t = any value of the continuous variable where t > 0

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The Exponential Distribution

Mean = $E(X) = 1/\lambda$

Var(X) $=(1/\lambda)^2$

Exponential Distribution Example

Worked Example 1

The interval, X seconds, between cars passing a point on a motorway follows an exponential distribution with probability density function

$$f(x) = \begin{cases} 2e^{-2x}, & x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

(i) State the mean and variance of X.

(ii) State the cumulative distribution function.

(iii) Calculate the probability that (give all answers to 3 significant figures):

a. The interval until the next car passes is between 1 and 2 seconds.

b. The interval until the next car passes is longer than 3 seconds.

c. The interval until the next car passes is less than 1.5 seconds.

https://courses.lumenlearning.com/introstats1/

Solution:

(i) This is an exponential distribution with parameter $\lambda = 2$. Therefore, the mean is $E(X) = \frac{1}{\lambda} = \frac{1}{2}$ and $\text{Var}(X) = \frac{1}{\lambda^2} = \frac{1}{4}$.

(ii) The cumulative distribution function is $F(x) = \begin{cases} 1 - e^{-2x}, & x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$

(iii) Each probability can be calculated using the cumulative distribution function.

a. We calculate $P(1 \leq X \leq 2)$ as follows:

$$\begin{aligned} P(1 \leq X \leq 2) &= P(X \leq 2) - P(X \leq 1) \\ &= F(2) - F(1) \\ &= (1 - e^{-4}) - (1 - e^{-2}) \\ &= e^{-2} - e^{-4} \\ &= 0.117. \end{aligned}$$

b. We calculate $P(X > 3)$ as follows:

$$\begin{aligned} P(X > 3) &= 1 - P(X \leq 3) \\ &= 1 - F(3) \\ &= 1 - (1 - e^{-6}) \\ &= e^{-6} \\ &= 0.00248. \end{aligned}$$

c. We calculate $P(X < 1.5)$ as follows:

$$\begin{aligned} P(X < 1.5) &= F(1.5) \\ &= 1 - e^{3} \\ &= 0.950. \end{aligned}$$

Exponential Distribution Example

Example: Customers arrive at the service counter at the rate of 15 per hour. What is the probability that the arrival time between consecutive customers is less than three minutes?

- The mean number of arrivals per hour is 15, so $\lambda = 15$
- Three minutes is .05 hours
- $P(\text{arrival time} < .05) = 1 - e^{-\lambda X} = 1 - e^{-(15)(.05)} =$ 0.5276
- So there is a 52.76% probability that the arrival time between successive customers is less than three minutes

The number of days ahead travelers purchase their airline tickets can be modeled by an exponential distribution with the average amount of time equal to 15 days. Find the probability that a traveler will purchase a ticket fewer than ten days in advance. How many days do half of all travelers wait?

- The mean amount of time equal to 15 days, so $\lambda^{-1} = 15$
- $P(X < 10) = 1 - e^{\left(-\frac{1}{15} * 10\right)}$
- $P(X < 10) = 1 - e^{\left(-\frac{2}{3}\right)}$
- $P(X < 10) \approx 1 - 0.5134 = 0.4866$
- **Result: The probability that a traveler purchases a ticket fewer than ten days in advance is 0.4866 (or 48.66%).**

Median Time (How long half of all travelers wait)

To determine the number of days that splits the population of travelers exactly in half, we calculate the median by setting the CDF equal to 0.50.

$$1 - e^{-\lambda x} = 0.50$$
$$\Rightarrow e^{-\lambda x} = 0.50$$
$$\Rightarrow -\lambda x = \ln(0.50)$$
$$\Rightarrow x = \ln(0.50) / (-\lambda)$$
$$\Rightarrow x = -0.6931 / (-1/15)$$
$$\Rightarrow x = 0.6931 * 15 \approx 10.40\ days$$

Result: Half of all travelers wait 10.40 days or fewer to purchase their ticket, while the remaining half purchase them more than 10.40 days in advance.

The time in hours required to repair a machine is exponentially distributed with perimeter $\lambda = 1/2$

(i) What is the probability that the repair time exceeds 2h,

(ii) What is the conditional probability that a repair takes at least 10h given that its duration exceeds 9h ?

The time in hours required to repair a machine is exponentially distributed with perimeter λ = 1/2
(i) What is the probability that the repair time exceeds 2h,
(ii) What is the conditional probability that a repair takes atleast 10h given that its duration exceeds 9h ?

Solution:
Given λ = 1/2
Let 'X' denotes the time to repair the machine.
The density function of X is given by,

$$f(x) = \lambda e^{-\lambda x} = \frac{1}{2} e^{\frac{-1}{2}x} \; ; x \geq 0$$

(i) P(the repair time exceeds 2h) = P(X > 2)

$$= \int_2^\infty \frac{1}{2} e^{\frac{-1}{2}x} \, dx$$

$$= \frac{-1}{2} \cdot 2 \left[e^{\frac{-1}{2}x} \right]_2^\infty = e^{-1} = 0.3679$$

(ii) The conditional probability that a repair takes at least 10h given that its duration exceeds 9h is given by,

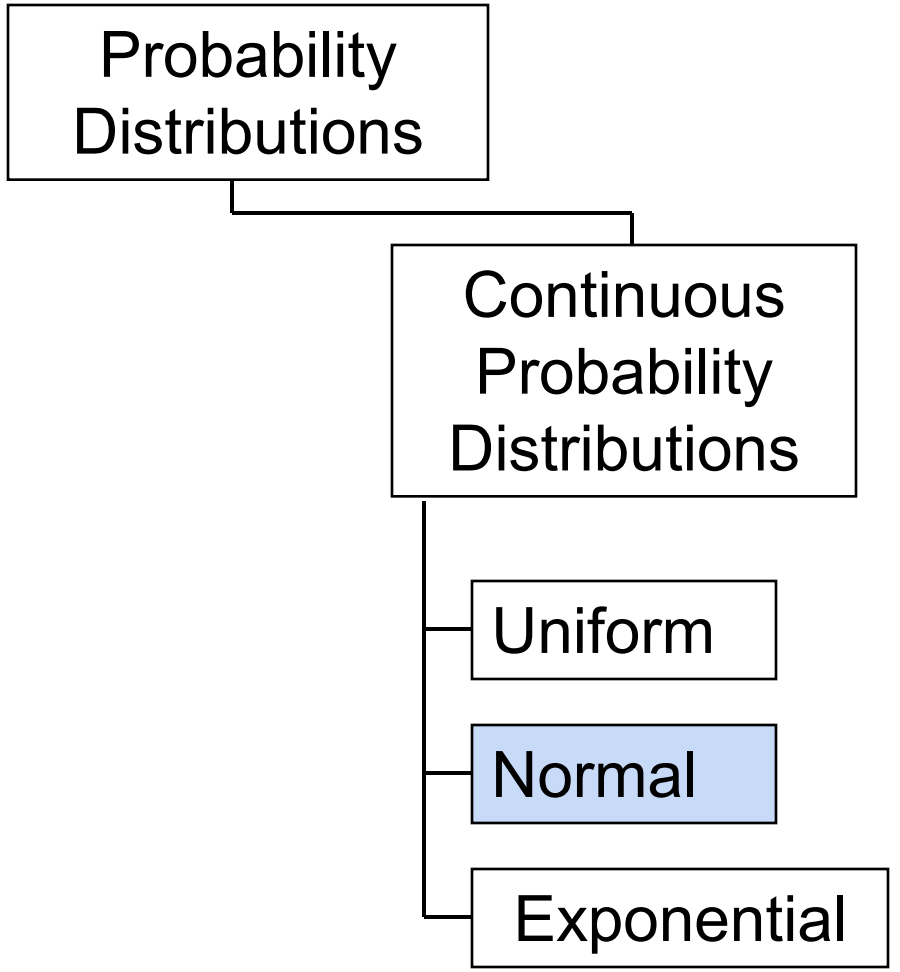
$$P(X \geq 10 \mid X > 9) = P(X \geq 9 + 1 \mid X > 9) = P(X > 1)$$

$$= \int_1^\infty \frac{1}{2} e^{\frac{-1}{2}x} \, dx = \frac{1}{2} \int_1^\infty e^{-\frac{1}{2}x} \, dx$$

$$= \frac{1}{2} \left[\frac{e^{-\frac{1}{2}x}}{-\frac{1}{2}} \right]_1^\infty = \frac{1}{2}(-2) \left[e^{-\frac{1}{2}x} \right]_1^\infty = (-1)\left[0 - e^{-\frac{1}{2}}\right] = e^{-\frac{1}{2}}$$

$$= 0.6065$$

The Normal Distribution



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Normal Distribution

This distribution is a continuous distribution and is a limiting case of Binomial Distribution when number of trials "n" is very large but $p\ and\ q$ are not so small.

A continuous RV X, is said to follow Normal distribution with parameters μ (mean) and σ^2 (Variance), if its pdf is

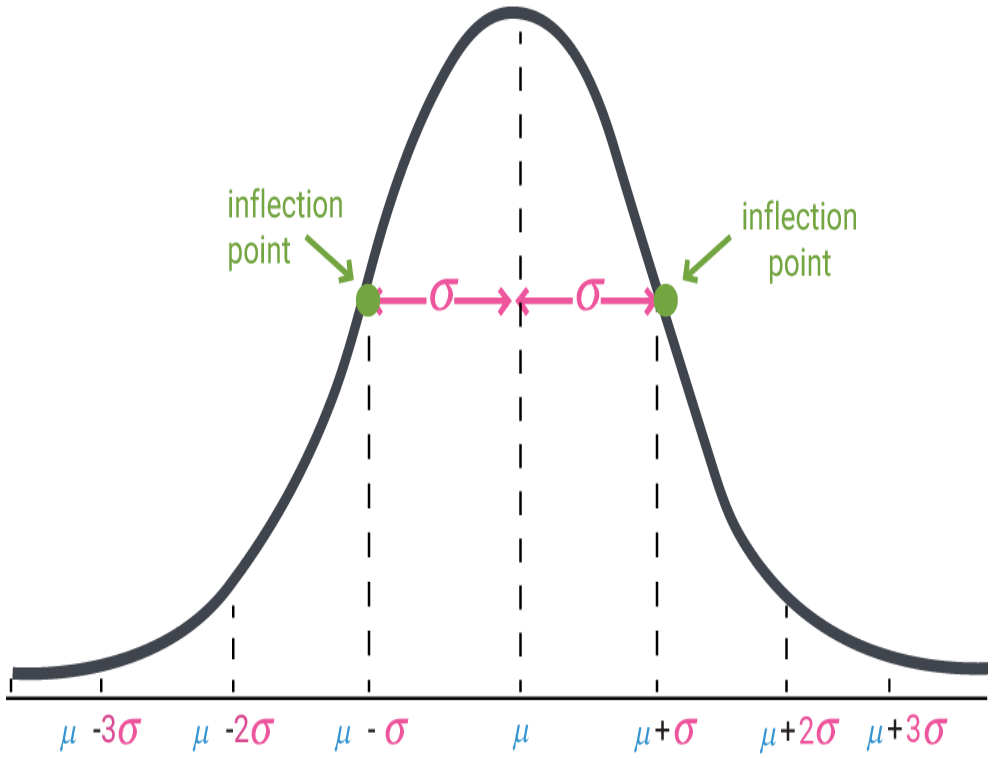
$$f(x,\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$-\infty < x < \infty, -\infty < \mu < \infty, \sigma > 0$$

Then, X is said to be normally distributed and denoted by $X \sim N(\mu,\sigma)$.

The function $f(x,\mu,\sigma)$ is symmetric about the line $x = \mu$.

The graph of the function is a bell shape curve and called as normal curve.



Standard Normal Variate

Let us do the following substitution in the pdf of Normal distribution

$$\frac{x-\mu}{\sigma} = z$$

Then, the pdf is reduced as

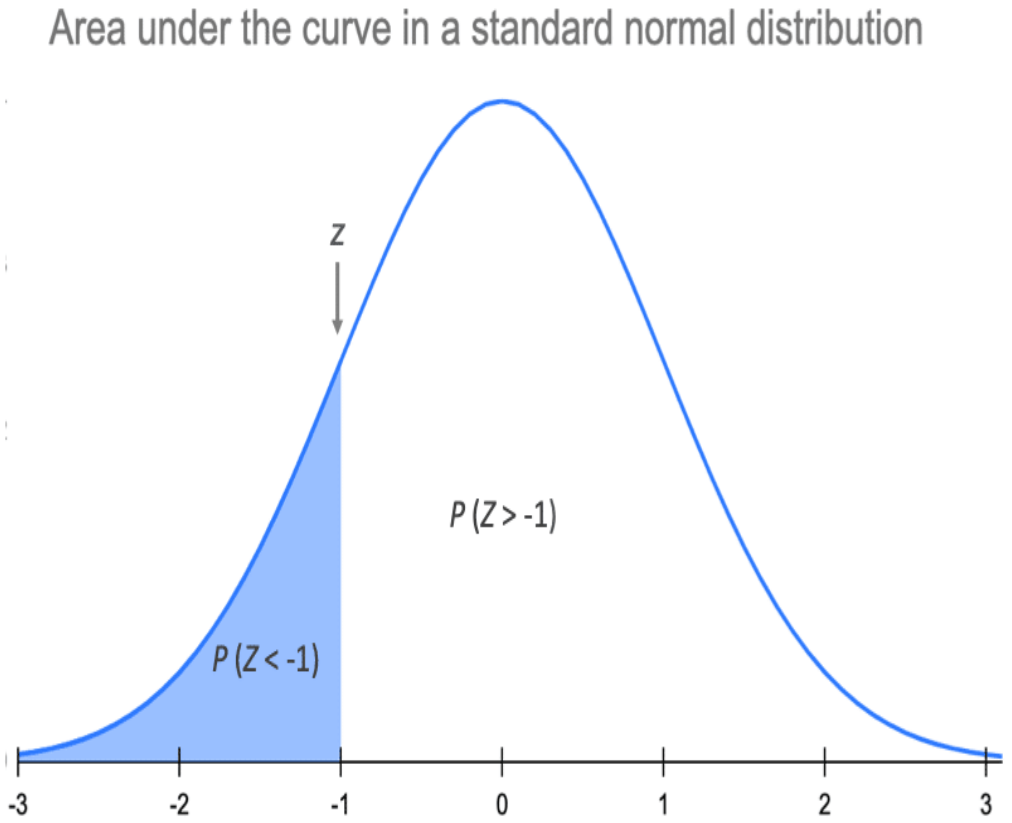
$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

$$-\infty < z < \infty$$

This density function is for the standard normal distribution.

For the standard normal distribution, the mean is zero and the variance is 1 and the corresponding random variable Z is called the standard normal variate.

The curve represented by this function is called standard normal curve and it is symmetrical about the line z = 0.



The standard normal variate is free from any parameter. This helps us to compute areas under the normal probability curve by making use of standard table.

The total area under the normal curve and above x – axis is 1. Also

$$P[-\infty < X < \infty] = 1, P[0 < X < \infty] = \frac{1}{2}$$

For standard normal variate

$$P[Z \leq 0] = P[Z \geq 0] = \frac{1}{2} = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-\frac{z^2}{2}} dz$$

Ex. On a final examination in Mathematics, the mean was 72, and the standard deviation was 15. Determine the standard scores of students receiving the grades (i) 60 (ii) 93 (iii) 72

Sol. Here $\mu = 72, \sigma = 15$. The standard normal variate is

$$z = \frac{x - \mu}{\sigma}$$

$$(i)\ z = \frac{x - \mu}{\sigma} = \frac{60 - 72}{15} = -0.8$$

$$(ii)\ z = \frac{x - \mu}{\sigma} = \frac{93 - 72}{15} = 1.4$$

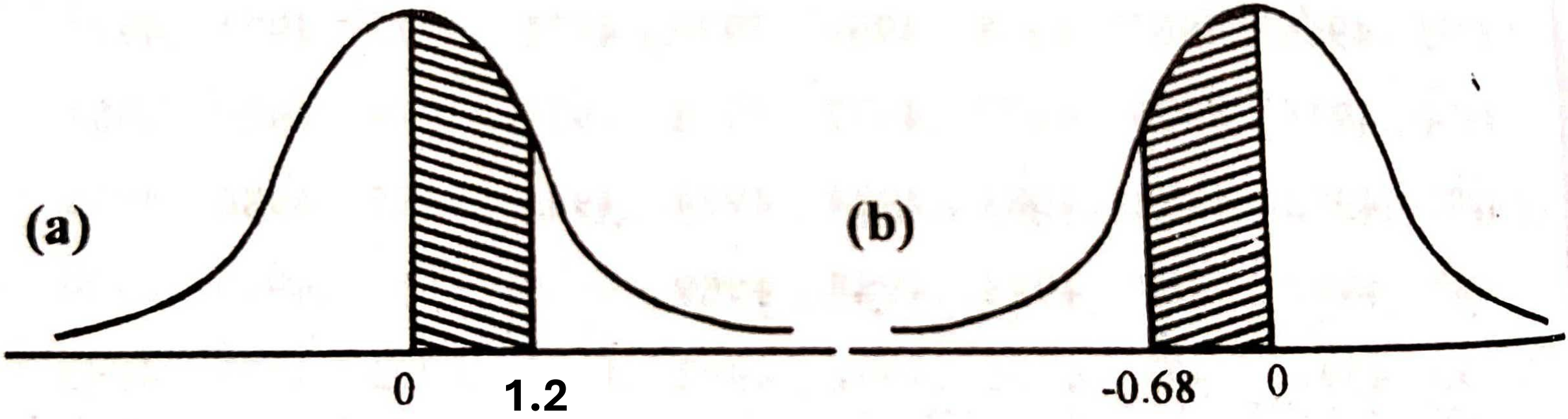
$$(iii)\ z = \frac{x - \mu}{\sigma} = \frac{72 - 72}{15} = 0$$

Ex. Find the area under the normal curve in each of the cases

(a) $z = 0$ and $z = 1.2$ (b) $z = 0$ and $z = -0.68$ (c) $z = -0.46$ and $z = 2.21$

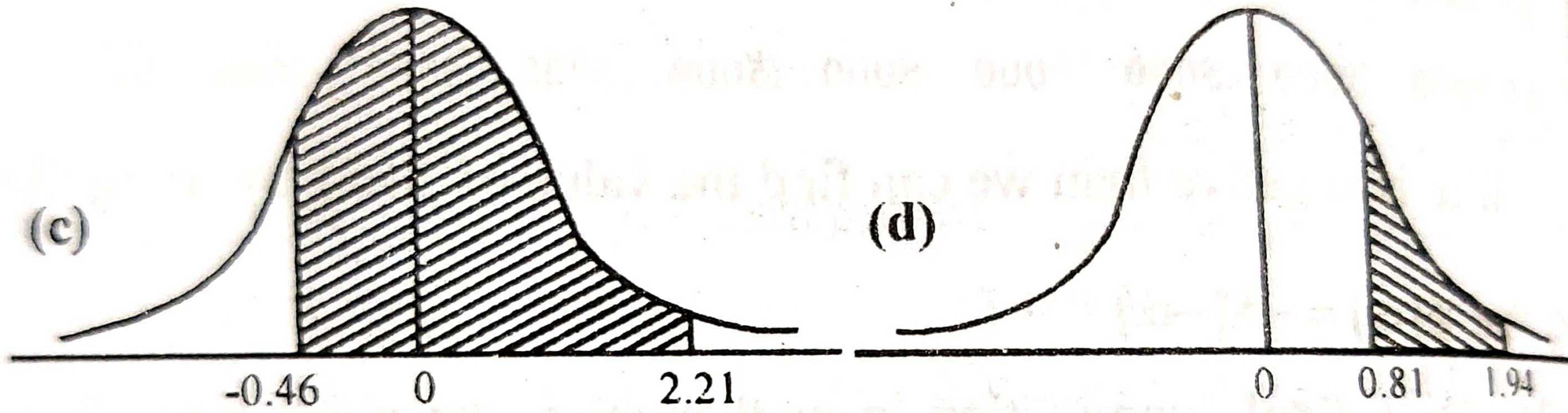
(d) $z = 0.81$ and $z = 1.94$ (e) To the left of $z = -0.6$ (f) To the right of $z = -0.28$

Sol. (a) Area between $z = 0$ and $z = 1.2$ and (b) Area between $z = 0$ and $z = -0.68$ is presented in the following figure



From the normal distribution table, the area between z = 0 and z = 1.2 is 0.3849.
As the curve is symmetric about z = 0, so the area between z = -0.68 and z = 0 is same as the area between z = 0 and z = 0.68. So, from the normal distribution table, the required area is 0.2517.

(c) $z = -0.46$ and $z = 2.21$ and (d) $z = 0.81$ and $z = 1.94$



The area between z = -0.46 and z = 2.21

= (Area between z = -0.46 and z = 0) + (Area between z = 0 and z = 2.21)

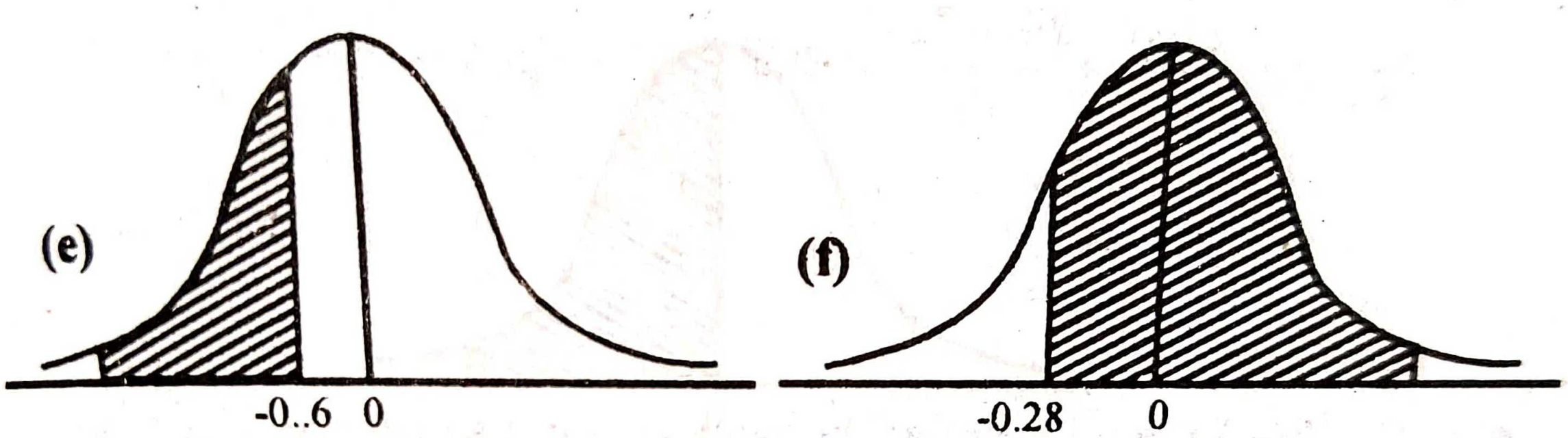
= (Area between z = 0 and z = 0.46) + (Area between z = 0 and z = 2.21)

= 0.1772 + 0.4864 = 0.6636

The area between z = 0.81 and z = 1.94

= (Area between z = 0 and z = 1.94) - (Area between z = 0 and z = 0.81)

= 0.4738 – 0.2910 = 0.1828



(e) Area to the left of z = -0.6

= (Area between $z = -\infty$ and z = 0) – (Area between z = -0.6 and z = 0)

= 0.5 – (Area between z = 0 and z = 0.6) = 0.5 – 0.2257 = 0.2743

(f) Required area = (Area between $z = -0.28$ and z = 0) + (Area between z = 0 and $z = \infty$)

= (Area between z = 0 and $z = 0.28$) + 0.5 = 0.1103 + 0.5 = 0.6103

General Procedure for Finding Probabilities

To find $P(a < X < b)$ when X is distributed normally:

- Draw the normal curve for the problem in terms of X
- Translate X-values to Z-values
- Use the Cumulative Normal Table

Ex. In a male population of 1000, the mean height is 68.16 inches and standard deviation is 3.2 inches. How many men may be more than 6 feet.

Sol. Here $\mu = 68.16\ inches, \sigma = 3.2\ inches,$

Male population = 1000, $x > 6\ ft = 72\ inches$

for standard normal variate

$$z = \frac{x - \mu}{\sigma} = \frac{72 - 68.16}{3.2} = 1.2$$

Required probability = Probability for the men more than 6 ft (72 inches)

$$= P[X > 72] = P[Z > 1.2]$$

= Area between z = 1.2 and z = ∞

= (Area between z = 0 and $z = \infty$) – (Area between z = 0 and z = 1.2)

= 0.5 – 0.3849 = 0.1151

Men more then 6 ft = $1000 * 0.1151$

$= 115.1 \approx 115$

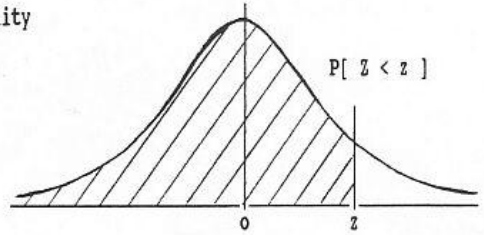
NAAC A+ GRADE WITH 3.28 SCORE

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability up to the standardised normal value z
i.e.

$$P[\ Z < z\ ] = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \exp(-\tfrac{1}{2}Z^2)\, dZ$$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5159	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7854
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8804	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9773	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9865	0.9868	0.9871	0.9874	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9980	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
z	3.00	3.10	3.20	3.30	3.40	3.50	3.60	3.70	3.80	3.90
P	0.9986	0.9990	0.9993	0.9995	0.9997	0.9998	0.9998	0.9999	0.9999	1.0000

TABLE OF AREAS

↓Z→	0	1	2	3	4	5	6	7	8	9
·0	·0000	·0040	·0080	·0120	·0160	·0199	·0239	·0279	·0319	·0359
·1	·0398	·0438	·0478	·0517	·0557	·0596	·0636	·0675	·0714	·0759
·2	·0793	·0832	·0871	·0910	·0948	·0987	·1026	·1064	·1103	·1141
·3	·1179	·1217	·1255	·1293	·1331	·1368	·1406	·1443	·1480	·1517
·4	·1554	·1591	·1628	·1664	·1700	·1736	·1772	·1808	·1844	·1879
·5	·1915	·1950	·1985	·2019	·2054	·2088	·2123	·2157	·2190	·2224
·6	·2257	·2291	·2324	·2357	·2389	·2422	·2454	·2486	·2517	·2549
·7	·2580	·2611	·2642	·2673	·2703	·2734	·2764	·2794	·2823	·2852
·8	·2881	·2910	·2939	·2967	·2995	·3023	·3051	·3078	·3106	·3133
·9	·3159	·3186	·3212	·3238	·3264	·3289	·3315	·3340	·3365	·3389
1·0	·3413	·3438	·3461	·3485	·3508	·3531	·3554	·3577	·3599	·3621
1·1	·3643	·3655	·3686	·3708	·3729	·3749	·3770	·3790	·3810	·3830
1·2	·3849	·3869	·3888	·3907	·3925	·3944	·3962	·3980	·3997	·4015
1·3	·4032	·4049	·4066	·4082	·4099	·4115	·4131	·4147	·4162	·4177
1·4	·4192	·4207	·4222	·4236	·4251	·4265	·4279	·4292	·4306	·4319
1·5	·4332	·4345	·4357	·4370	·4382	·4394	·4406	·4418	·4429	·4441

TABLE OF AREAS

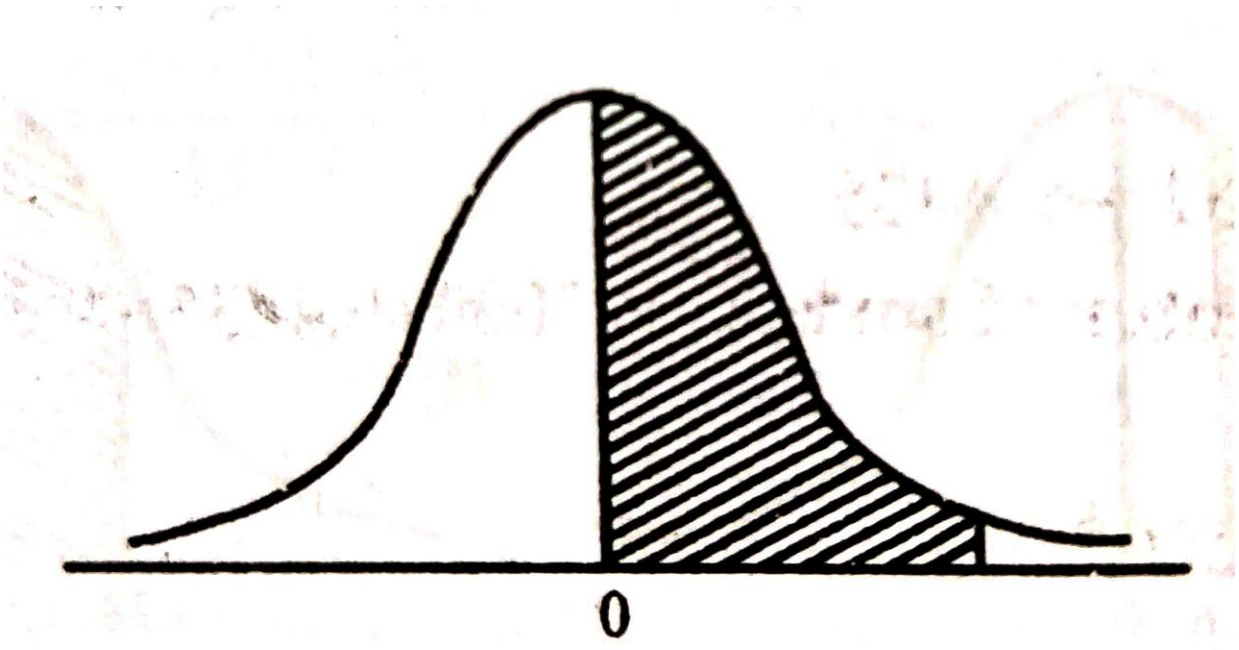
↓z→	0	1	2	3	4	5	6	7	8	9
1·6	·4452	·4463	·4474	·4484	·4495	·4505	·4515	·4525	·4535	·4545
1·7	·4554	·4564	·4573	·4582	·4591	·4599	·4608	·4616	·4625	·4633
1·8	·4641	·4649	·4656	·4664	·4671	·4678	·4686	·4693	·4699	·4706
1·9	·4713	·4719	·4726	·4732	·4738	·4744	·4750	·4756	·4761	·4767
2·0	·4772	·4778	·4783	·4788	·4793	·4798	·4803	·4808	·4812	·4817
2·1	·4821	·4826	·4830	·4834	·4838	·4842	·4846	·4850	·4854	·4857
2·2	·4861	·4864	·4868	·4871	·4875	·4678	·4881	·4884	·4887	·4890
2·3	·4893	·4896	·4898	·4901	·4904	·4906	·4909	·4911	·4913	·4916
2·4	·4918	·4920	·4922	·4925	·4927	·4929	·4931	·4932	·4934	·4936
2·5	·4938	·4940	·4941	·4943	·4945	·4946	·4948	·4959	·4951	·4952
2·6	·4953	·4955	·4956	·4957	·4959	·1960	·4961	·4962	·4963	·4964
2·7	·4965	·4966	·4967	·4968	·4969	·4970	·4971	·4972	·4973	·4974
2·8	·4974	·4975	·4976	·4977	·4977	·4978	·4979	·4979	·4980	·4981
2·9	·4981	·4982	·4982	·4983	·4984	·4984	·4985	·4985	·4986	·4986
3·0	·4987	·4987	·4987	·4988	·4988	·4989	·4989	·4989	·4990	·4990
3·1	·4990	·4991	·4991	·4991	·4992	·4992	·4992	·4992	·4993	·4993
3·2	·4993	·4993	·4994	·4994	·4994	·4994	·4994	·4995	·4995	·4995
3·3	·4995	·4995	·4995	·4996	·4996	·4996	·4996	·4996	·4996	·4997
3·4	·4997	·4997	·4997	·4997	·4997	·4997	·4997	·4997	·4997	·4998
3·5	·4998	·4998	·4998	·4998	·4998	·4998	·4998	·4998	·4998	·4998
3·6	·4998	·4998	·4999	·4999	·4999	·4999	·4999	·4999	·4999	·4999
3·7	·4999	·4999	·4999	·4999	·4999	·4999	·4999	·4999	·4999	·4999
3·9	·5000	·5000	·5000	·5000	·5000	·5000	·5000	·5000	·5000	·5000

Ex. Students of a class were given an aptitude test. Their marks were found to be normally distributed with mean 60 and standard deviation 5. What percentage of students scored more than 60 marks.

Sol. Here, $x = 60, \mu = 60 \; and \; \sigma = 5$. So, for standard normal variate

$$z = \frac{x - \mu}{\sigma} = \frac{60 - 60}{5} = 0$$

$If \; x > 60$ then $z > 0$. The area lying to the right of z = 0 is 0.5.



Finding the X value for a Known Probability

- Steps to find the X value for a known probability:
  1. Find the Z value for the known probability
  2. Convert to X units using the formula:

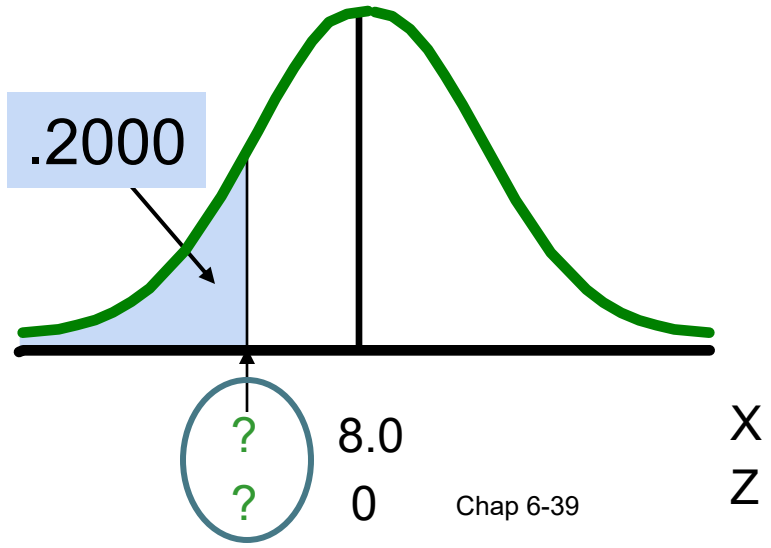
$$X = \mu + Z\sigma$$

Finding the X value for a Known Probability

(continued)

Example:

- Suppose X is normal with mean 8.0 and standard deviation 5.0.
- Now find the X value so that only 20% of all values are below this X



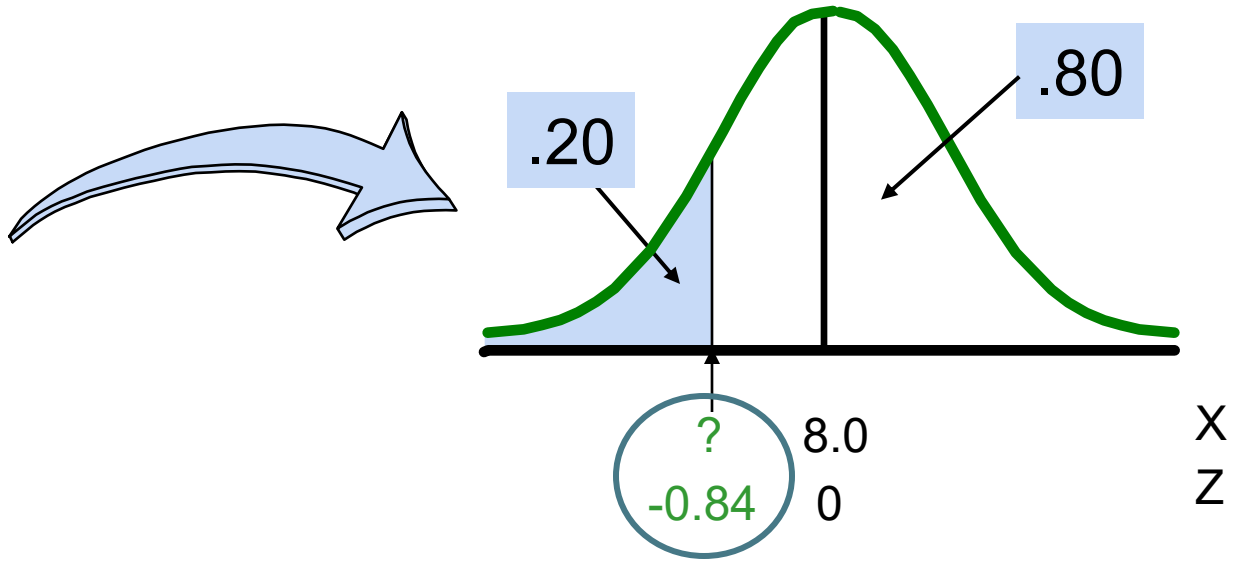
Find the Z value for 20% in the Lower Tail

1. Find the Z value for the known probability

Standardized Normal Probability Table (Portion)

z	F(z)
.82	.7939
.83	.7967
.84	.7995
.85	.8023

- 20% area in the lower tail is consistent with a Z value of -0.84



Finding the X value

2. Convert to X units using the formula:

$$X = \mu + Z\sigma$$

$$= 8.0 + (-0.84)5.0$$

$$= 3.80$$

So 20% of the values from a distribution with mean 8.0 and standard deviation 5.0 are less than 3.80

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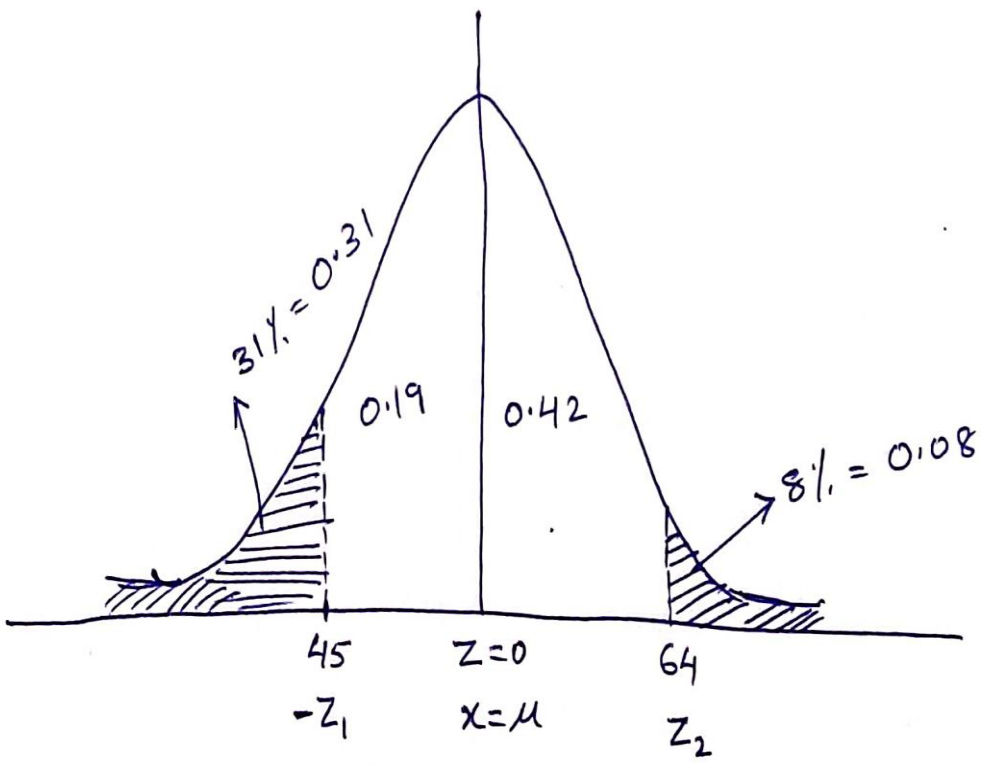
Ex. In a normal distribution, 31% items are under 45 and 8% are over 64. Find the mean and the standard deviation of the distribution.

Sol. As $P[X < 45] = 31\% = 0.31$ and it must be on LHS region otherwise the probability be (0.5 + 0.31) = 0.81.

$P[X > 64] = 8\% = 0.08$ and it must be on RHS region otherwise the probability be (0.5 + 0.08) = 0.58.

For the standard normal variate

$$z = \frac{x - \mu}{\sigma}$$



For $x = 45$

$$-z_1 = \frac{45 - \mu}{\sigma} \quad \ldots(1)$$

The area between $z = -z_1$ and z = 0

= The area between z = 0 and $z = z_1$

= 0.5 – 0.31 = 0.19

As from the normal distribution table, the nearest value for the area 0.19 is 0.1915 and it is true for $\phi(0 < z < 0.5) = 0.1915$. So,

$$z_1 = 0.5$$

From (1),

$$-0.5 = \frac{45 - \mu}{\sigma} \quad \ldots(2)$$

For $x = 64$

$$z_2 = \frac{64 - \mu}{\sigma} \quad \ldots(3)$$

The area between z = 0 and $z = z_2$

= (The area between z = 0 and $z = \infty$) – (The area between $z = z_2$ and $z = \infty$)

= 0.5 – 0.08 = 0.42

As from the normal distribution table, the nearest value for the area 0.42 is 0.427 and it is true for $\phi(0 < z < 1.41) = 0.427$. So,

$$z_2 = 1.41$$

From (3),

$$1.41 = \frac{45 - \mu}{\sigma} \quad \ldots(4)$$

(4)/(2) we have

$$-\frac{1.41}{0.5} = \frac{64-\mu}{45-\mu}$$

$$0.5(64-\mu) = -1.41(45-\mu)$$

$$\mu = 49.97 \approx 50$$

Putting this value in (4) and after calculations we get

$$\sigma = 9.929 \approx 10$$

References

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